AJITHRAM A S

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Professional Summary

M.Sc. Integrated Data Science graduate with hands-on internship experience in software testing, backend development, and cloud platform engineering. Skilled in API testing and automation using Postman and Newman, Python compatibility and framework testing, Kubernetes-based deployment validation, and CI/CD pipeline integration. Experienced in building backend systems with FastAPI, Flask, and Django, and in developing machine learning models for real-world classification tasks. Seeking entry-level roles in Software Testing, QA Automation, Backend Development, or Data Science.

Education

Coimbatore Institute of Technology **Sep 2021 – May 2026**
M.Sc. (Integrated) Data Science *CGPA: 7.92*

Tagore Matric Higher Secondary School **May 2021**
Higher Secondary Certificate (HSC) *88.53%*

Tagore Matric Higher Secondary School **Mar 2019**
Secondary School Leaving Certificate (SSLC) *93.2%*

Technical Skills

Programming: Python, Java
Backend Frameworks: FastAPI, Flask, Django, Express.js
Testing & QA: Manual Testing, UI Testing, API Testing, Integration Testing, Compatibility Testing
Test Automation: Postman, Newman, GitLab CI/CD, GitHub Actions
Cloud & DevOps: AWS, Kubernetes, Docker, AWS ECR, Helm, Prometheus, Grafana, ElasticSearch
Databases: MySQL, MongoDB, PostgreSQL
Data Science & ML: Pandas, Scikit-learn, TensorFlow, Keras, XGBoost, CNN, NLP, EDA, Data Analysis, Transfer Learning
Other Tools: Power BI, MS Excel, DBeaver, Streamlit, Dialogflow, FusionAuth

Professional Experience

Software Testing Intern – Platform Engineer **Dec 2025 – Apr 2026**
Dilligen Technologies Pvt. Ltd. *Salem / Remote*

Nimbuz Cloud Platform Testing and Automation

- Validated end-to-end cloud application deployment workflows on a Kubernetes-based platform, covering environment configuration, tool installation, deployment execution, and lifecycle state monitoring (Starting, Running, Failed).
- Performed compatibility and integration testing for Flask, Django, and FastAPI applications across Python versions 3.7–3.13, identifying deployment failures and dependency conflicts across framework versions.
- Designed and executed functional, UI, integration, and deployment test cases covering tool creation, configuration validation, error handling, and application lifecycle management.
- Automated API validation using Postman and Newman, building test suites for request validation, response body checks, HTTP status code verification, and error-response handling, integrated into GitLab CI/CD pipelines.
- Analyzed application logs, investigated deployment errors, and documented defects for root-cause analysis; performed regression testing post-fix to verify resolution.

Software Developer Intern – Backend Developer **Jul 2024 – Jan 2025**
Dilligen Technologies Pvt. Ltd. *Salem / Remote*

Enhancing Cloud Efficiency with Kubernetes (Nimbuz Cloud)

- Built a modular Kubernetes-based cloud management platform, reducing setup time by 60% for SMEs with limited cloud infrastructure expertise.
- Deployed and scaled 25+ containerized services with persistent storage and real-time monitoring using Prometheus and Grafana.
- Automated CI/CD pipelines using GitHub Actions, streamlining release cycles and enabling continuous integration for containerized application deployments.
- Integrated FusionAuth with Role-Based Access Control (RBAC), ensuring secure authentication and authorization across the cloud platform.
- Validated platform APIs using Postman; worked with AWS ECR for Docker image management and Helm for Kubernetes application packaging.

Projects

Healthcare Image Classification using CNNs | *Python, TensorFlow, Keras, Grad-CAM*

- Built and trained a CNN model on 5,000+ chest X-ray and CT scan images, achieving 93% accuracy in COVID-19 detection from medical imaging data.
- Improved model generalization by 12% on validation data using data augmentation and dropout regularization techniques.
- Applied Grad-CAM visualizations and confusion matrices to enhance model interpretability and support clinical decision-making workflows.

Food Delivery Chatbot | *Dialogflow, Python, FastAPI, MySQL*

- Architected an end-to-end food delivery chatbot using Dialogflow and FastAPI, enabling real-time order processing and natural language user interaction.
- Designed conversational flows by defining intents, entities, and contexts in Dialogflow to ensure accurate natural language understanding.
- Engineered a scalable FastAPI backend to manage chatbot requests, process food orders, and dynamically interact with a MySQL database.

Jewelry Class Prediction | *Python, Pandas, Scikit-learn, VGG16, XGBoost*

- Performed EDA and data cleaning on a structured jewelry dataset with 8+ features including purity, category, and weight band to ensure high data quality.
- Resolved a 34% accuracy drop caused by feature removal by integrating deep learning image features extracted using the pretrained VGG16 model.
- Combined visual and tabular features to train an XGBoost classifier, boosting final model accuracy from 65% to 93%.

Certifications & Achievements

- National Conference on Machine Learning and Artificial Intelligence – IIM Bangalore
- NPTEL – Cyber Security and Privacy (Oct 2023)
- QNickle Diploma – Quantum Computing and Programming (Apr 2025)